

Source Code

1. Adding a New Task

```
function addTask(){  
  
    const text = taskInput.value.trim();  
  
    if(text === ""){  
        alert("Please enter a task");  
        return;  
    }  
  
    tasks.push({  
        text:text,  
        category:category.value,  
        priority:priority.value,  
        dueDate:dueDate.value,  
        completed:false,  
        created:new Date().toLocaleString()  
    });  
  
    saveTasks();  
    renderTasks();  
  
    taskInput.value = "";  
}
```

Purpose

This function creates a new task, stores its details, and updates the interface.

2. Saving Tasks in Local Storage

```
function saveTasks(){  
    localStorage.setItem(  
        "tasks",  
        JSON.stringify(tasks)  
    );  
}
```

Purpose

Stores task data in the browser so that tasks remain available after refreshing the page.

3. Rendering Tasks Dynamically

```
function renderTasks(){

  taskList.innerHTML = "";

  tasks.forEach((task,index)=>{

    const li =
    document.createElement("li");

    li.innerHTML = `
      <div>
        <h3>${task.text}</h3>
        <p>📁 ${task.category}</p>
        <p>★ ${task.priority}</p>
      </div>
    `;

    taskList.appendChild(li);
  });
}
```

Purpose

Creates task elements dynamically using DOM manipulation.

4. Search Functionality

```
searchInput.addEventListener(
  "keyup",
  ()=>{

    const filter =
    searchInput.value.toLowerCase();

    document
    .querySelectorAll("li")
```

```

        .forEach(task=>{

            task.style.display =
            task.innerText
            .toLowerCase()
            .includes(filter)
            ? "flex"
            : "none";

        });
    });

```

Purpose

Allows users to search and filter tasks in real time.

5. Dark Mode Feature

```

themeBtn.onclick = ()=>{

    document.body
    .classList
    .toggle("dark");

    themeBtn.innerHTML =
    document.body
    .classList
    .contains("dark")
    ? "☀ Light Mode"
    : "🌙 Dark Mode";

};

```

Purpose

Switches between light and dark themes.

6. Export Tasks Feature

```

document
.getElementById("exportBtn")
.onclick = ()=>{

    let content =

```

```

        tasks.map(task =>
            `${task.text}
            Category: ${task.category}
            Priority: ${task.priority}

        `).join("\n");

        const blob =
        new Blob(
            [content],
            {type:"text/plain"}
        );

        const link =
        document.createElement("a");

        link.href =
        URL.createObjectURL(blob);

        link.download =
        "tasks.txt";

        link.click();
    };

```

Purpose

Exports all tasks into a downloadable text file.

7. Voice Input Feature

```

const recognition =
new(
    window.SpeechRecognition ||
    window.webkitSpeechRecognition
)();

document
.getElementById("micBtn")
.onclick = ()=>{

    recognition.start();
};

```

```
recognition.onresult =  
(event)=>{  
  
    taskInput.value =  
    event.results[0][0]  
    .transcript;  
};
```

Purpose

Allows users to add tasks using voice commands through the Web Speech API.